GROUP ASSIGNMENT 1

LINEAR REGRESSION: HOUSING PRICE PREDICTION

Google Colab Notebook: BUS310A_GroupAssignment1_Group5.ipynb

https://colab.research.google.com/drive/1P3Z9vHjXhTeoVcFjcrOp9x9-hRCdzfmM?usp=sharing

BUSA 310A Business Analytics III: Predictive and Prescriptive Business Analytics

Professor Yanchao Yang | Spring 2025 | March 9, 2025

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Abstract

The housing market is a key driver of economic activity, shaping investment strategies and policy decisions. While numerous methods exist for predicting housing prices, balancing accuracy, complexity, and interpretability remains a challenge. This study applies Ordinary Least Squares (OLS) regression to predict home prices in Ames, Iowa (2006–2010), comparing two models: a streamlined approach using the most influential predictors and a comprehensive model incorporating all statistically significant variables. Our results reveal that a simpler model focusing on key predictors achieves lower Mean Squared Error (MSE), offering more stable and reliable price estimates. In contrast, the comprehensive model provides greater explanatory power but at the cost of increased prediction error. This trade-off underscores the importance of model selection based on the specific needs of real estate professionals—whether prioritizing precise predictions or deeper insights into price determinants. The findings reinforce the utility of regression-based approaches for real estate valuation, offering a data-driven framework for market analysis and decision-making.

Keywords: *Housing Price Prediction, Linear Regression (OLS), Real Estate Valuation, Predictive Modeling, Feature Selection in Regression*

I. INTRODUCTION

Predicting housing prices is a critical area of analysis in real estate, influencing decisions made by homebuyers, sellers, investors, and policymakers. A home's value is determined by a complex interplay of physical characteristics, location, and market conditions, making it essential to develop accurate models for price estimation. Regression analysis, particularly linear regression, provides a powerful statistical approach to understanding these relationships and forecasting property prices. By identifying key factors that drive home values, predictive models can help stakeholders make more informed financial and investment decisions.

This study applies linear regression techniques to predict housing prices for residential properties sold in Ames, Iowa, between 2006 and 2010. The dataset consists of 81 variables, capturing information about property size, condition, neighborhood, sale details, and unique features such as pools, fences, and sheds. Our primary goals are:

- To identify the most influential factors affecting house prices.
- To compare multiple regression models in terms of predictive performance using Mean Squared Error (MSE) and R-squared (R^2).
- To assess how well linear regression can model housing prices and provide practical insights into real estate valuation.

To achieve these objectives, we preprocess the dataset by handling missing values, encoding categorical variables, and selecting the most relevant predictors. We then train and evaluate two regression models, comparing their effectiveness in predicting house prices. The insights gained from this analysis can provide valuable guidance for real estate professionals, homebuyers, and investors looking to make data-driven decisions.

1.1. Literature Review

Accurate housing price prediction is a cornerstone of real estate market analysis, impacting investment decisions, policy planning, and financial forecasting. Regression-based models have been widely studied as effective tools for understanding the relationships between housing attributes and market prices. Shukla (2024) highlights the efficacy of Multiple Linear Regression (MLR) in predicting home prices, emphasizing that proper data preprocessing—such as handling missing values and encoding categorical variables—is essential for improving model accuracy. The study further suggests that incorporating structured regression models can significantly enhance the predictive power of housing price estimations, providing a strong foundation for data-driven decision-making in the real estate industry.

Building on this, Fu (2024) compares the performance of Linear Regression and Random Forest models, concluding that while ensemble learning techniques like Random Forest yield higher predictive accuracy, linear regression remains a strong baseline for understanding price determinants. The study underscores the importance of building classification and living area size as the most influential predictors of home prices—aligning with our findings that Above-Grade Living Area (GrLivArea) and Overall Quality (OverallQual) exhibit the strongest correlations with SalePrice. This comparison of regression models further supports our decision to employ linear regression in our study, as it offers transparency and interpretability in evaluating housing market trends.

Beyond individual property characteristics, macroeconomic factors also play a crucial role in shaping housing prices. Dotsis, Petris, and Psychoyios (2023) analyze 150 years of historical housing market crashes, finding that deviations in the rent-to-price ratio and fluctuations in short-term interest rates serve as strong indicators of price downturns. Their study highlights how external economic forces influence real estate values, reinforcing the need to account for broader market conditions when modeling housing prices. Additionally, Møller et al. (2023) examine housing price predictability in different market conditions, revealing that search intensity and buyer behavior significantly affect price stability. These studies emphasize the dynamic nature of housing markets and suggest that while regression models can effectively capture structural attributes, incorporating macroeconomic trends could further enhance predictive accuracy.

1.2. Methodology & Key Findings

To answer our research question, we develop and test two regression models. Model 1 includes a smaller set of highly correlated variables, ensuring a streamlined approach that prioritizes predictive accuracy. Model 2 incorporates a broader range of features, capturing more complex relationships within the dataset. We evaluate both models using MSE and R^2 to determine their respective strengths in price estimation.

Our results show that while Model 2 exhibits a higher R^2 value, indicating a stronger overall fit to the dataset, Model 1 achieves a lower MSE, suggesting it provides more stable and accurate price predictions. This trade-off highlights the importance of balancing model complexity with predictive reliability. The key implication of our findings is that a focused selection of high-impact variables, rather than an overly complex model, may lead to better price predictions in real estate valuation.

1.3. Implications & Structure of the Paper

These findings have direct applications for real estate professionals, financial analysts, and homebuyers. By understanding which housing attributes most strongly influence price, stakeholders can

make more data-driven decisions in pricing strategies, investment choices, and market evaluations. Additionally, our study reinforces the role of regression-based models in real estate forecasting while highlighting the need to consider external economic influences.

This paper is structured as follows:

- The Data Section outlines the dataset, preprocessing steps, and feature selection techniques.
- The Methodology Section describes the regression models used and their evaluation criteria.
- The Results Section presents findings from model testing and performance comparison.
- Finally, the Conclusion Section discusses key takeaways, study limitations, and directions for future research.

II. DATA SECTION

The dataset used in this study consists of 2,919 observations of residential properties sold in Ames, Iowa, between 2006 and 2010. It initially contained 81 variables, encompassing both numerical and categorical attributes that describe key property characteristics such as lot size, structural details, room counts, sale conditions, and additional features like pools, fences, and sheds. Given the complexity of real estate pricing, careful preprocessing was necessary to ensure the data's quality and suitability for regression analysis.

To ensure data integrity, missing values were handled systematically. Numerical variables were imputed using their mean values, while categorical variables were imputed using the most frequent category within each respective column. This approach minimized bias while maintaining consistency across observations.

Additionally, some numerical variables that serve as categorical indicators—such as MSSubClass (building class), MoSold (month sold), and YearBuilt (construction year)—were converted into dummy variables to enhance interpretability in the regression models. By treating YrSold and MoSold as categorical rather than continuous variables, we ensured that our model captures year-to-year and month-to-month market variations more effectively. This approach aligns with real estate market trends, where housing prices are influenced by seasonal fluctuations and broader economic conditions rather than a strictly linear progression over time.

After preprocessing, the dataset expanded as categorical variables were transformed into dummy variables using one-hot encoding, resulting in a final set of 34 numerical variables, with a significantly larger total number of predictors after encoding.

The target variable, SalePrice, represents the final transaction price of each property. Its distribution is right-skewed, with values ranging from $34,900 to $755,000, a mean of $180,052.86, and a standard deviation of $57,381.56. This wide variation reflects the diverse housing market in Ames, Iowa, where factors such as home quality, size, and location significantly impact property values.

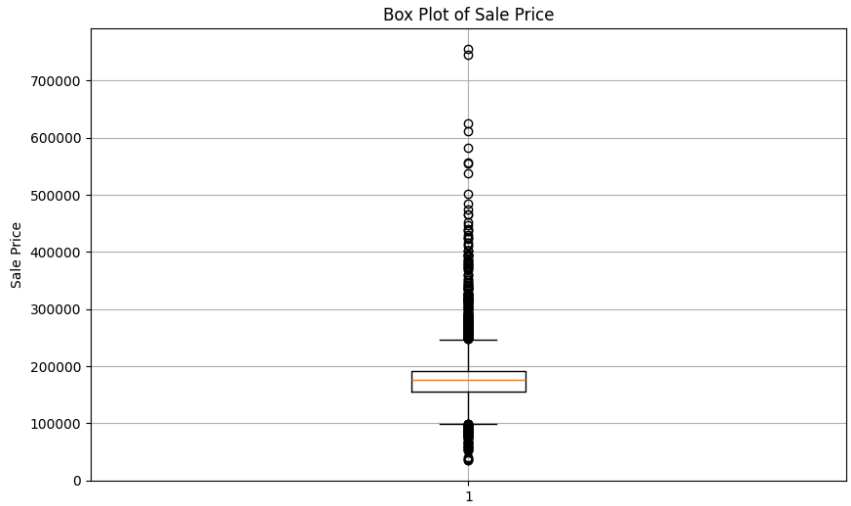


Figure 1: Box Plot of Sale Price

The histogram below visualizes the distribution of SalePrice, revealing a right-skewed pattern, meaning that a small number of high-value homes raise the overall mean. This indicates the potential need for log transformation to normalize the distribution for improved model performance.

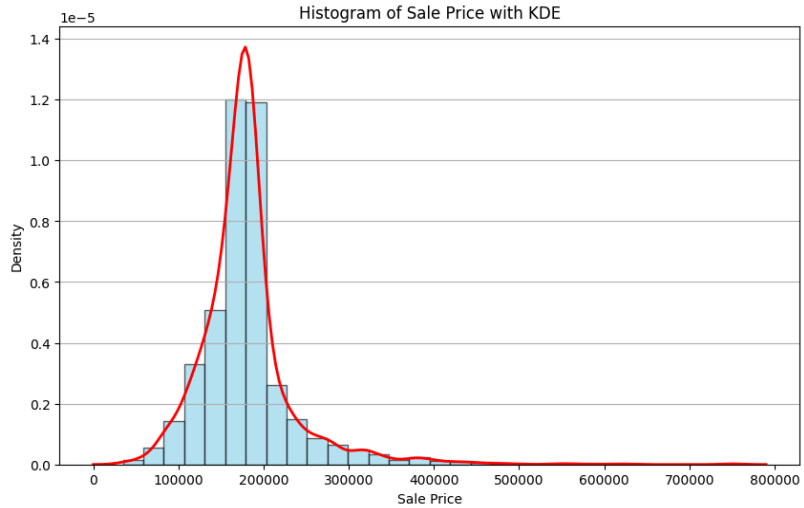


Figure 2: Histogram of SalePrice with KDE

To determine the most influential predictors, a correlation analysis was conducted. The results indicate that the strongest correlations with SalePrice are:

- Above-Grade Living Area (GrLivArea) (0.588)
- Overall Quality (OverallQual) (0.551)
- Total Rooms Above Ground (TotRmsAbvGrd) (0.470)

These findings align with real estate valuation principles, where factors such as living space, home quality, and total room count significantly impact property values.

```
Top 10 features most correlated with SalePrice:
GrLivArea       0.588010
OverallQual     0.550911
TotRmsAbvGrd    0.469800
GarageCars      0.469249
GarageArea      0.464809
1stFlrSF        0.462865
TotalBsmtSF     0.453212
FullBath        0.433711
YearBuilt       0.362066
MasVnrArea      0.353953
Name: SalePrice, dtype: float64
```

Figure 3: Top 10 Numerical Features Most Correlated with SalePrice

This is further supported by Fu (2024), who emphasizes that building classification and living area size are among the most critical variables influencing home prices. It aligns with our findings that Above-Grade Living Area (GrLivArea) and Overall Quality (OverallQual) exhibited the highest correlations with SalePrice. The study's comparative analysis of regression models reinforces our decision to use linear regression for predictive modeling.

The Overall Quality of a home is a crucial factor influencing its sale price, as higher quality homes typically command higher market values. The box plot below visually represents the relationship between Overall Quality (X-axis) and SalePrice (Y-axis). As expected, homes with a higher quality rating tend to have significantly higher sale prices, with a clear upward trend in median values as quality improves. Lower quality homes (rated 1-4) generally fall within a lower price range, while homes rated 8, 9, and 10 exhibit both higher median prices and greater variability in sale prices. The presence of outliers suggests that certain homes, despite having lower quality ratings, may have sold at higher prices due to other influential factors such as location or additional features. This visualization reinforces the importance of Overall Quality as a key predictor in our housing price prediction model.

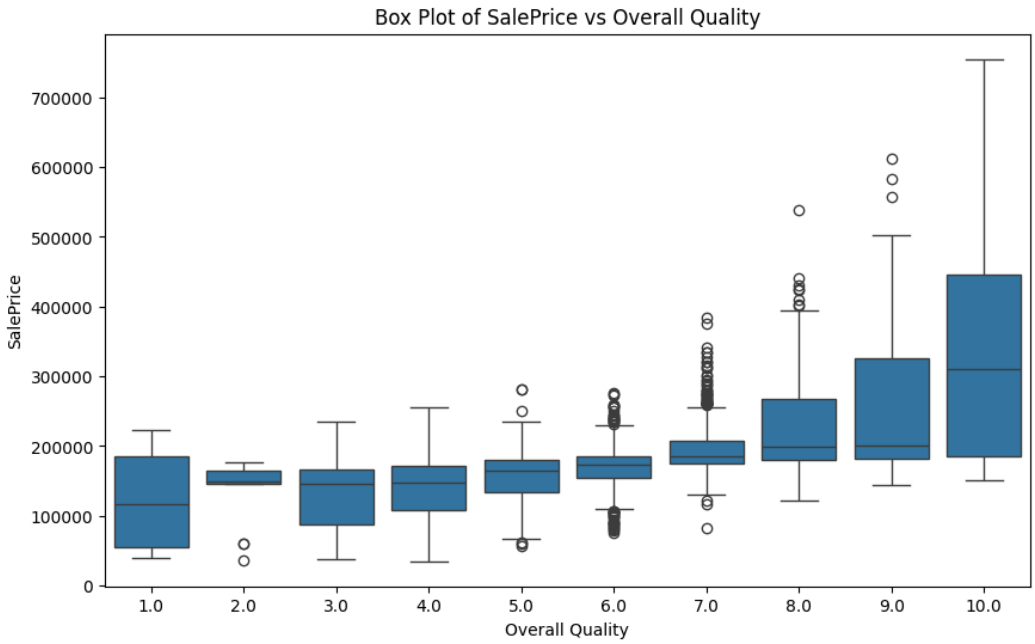


Figure 4: Box Plot of SalePrice vs. Overall Quality

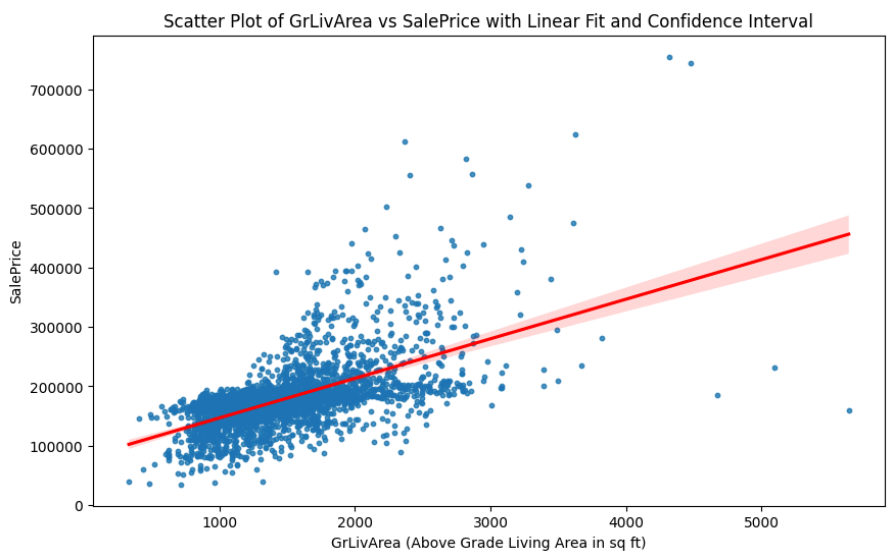


Figure 5: Scatter Plot of GrLivArea vs. SalePrice

This scatter plot demonstrates a positive linear relationship between GrLivArea and SalePrice, suggesting that larger homes tend to be sold at higher prices. However, there are a few outliers where properties with significantly high square footage have unexpectedly lower prices, which may indicate poor condition, undesirable locations, or external market factors.

In this study, Model 1 and Model 2 take different approaches to predictor selection, each with its own advantages and trade-offs:

- Model 1 prioritizes interpretability and simplicity by selecting only the top predictors based on their strongest correlations with SalePrice. The final variables—GrLivArea, OverallQual, TotRmsAbvGrd, TotalBsmtSF, YearBuilt, and MasVnrArea—were chosen to ensure a streamlined model that captures the most influential factors while minimizing complexity and reducing the risk of overfitting.
- In contrast, Model 2 adopts a more comprehensive approach by incorporating all statistically significant predictors (p-value < 0.05), resulting in 38 variables. While this model may better capture the nuances of housing price determinants, the increased number of features raises the risk of overfitting, potentially making it less generalizable to unseen data. The comparison between these models allows us to evaluate the trade-off between model complexity and predictive accuracy, providing insights into the most effective way to forecast housing prices.

III. METHODS SECTION

The methods section of this paper will explain how the study utilizes two models to examine the factors influencing housing prices, with the construction process for each model outlined below.

For Model 1, we selected the top variables with the strongest correlation to the sale price. Those variables are below in descending order.

Top Variables:

1. GrLivArea 0.588010
2. OverallQual 0.550911
3. TotRmsAbvGrd 0.469800
4. GarageCars 0.469249
5. GarageArea 0.464809
6. 1stFlrSF 0.462865
7. TotalBsmtSF 0.453212
8. FullBath 0.433711
9. YearBuilt 0.362066
10. MasVnrArea 0.353953

After discovering the top variables that have the most correlation with sale price, a heat map figure was created to illustrate these correlations. Below, this figure demonstrates that there are variables that are highly correlated with each other. For example, it is shown that “GarageCars” and “GarageArea” have a very high correlation of 0.89. Another example of highly correlated categories are “TotRmsAbvGrd” and “GrLivArea” which have a 0.81 correlation. Next, we determined that categories

were insignificant if they have a p_value>= 0.05. We removed categories with a p_value that was >= 0.05. With this being said, these methods moved us towards our final predictors which were GrLivArea, OverallQual, TotRmsAbvGrd, GarageArea, TotalBsmtSF, and MasVnrArea. After gathering our final predictors a regression was run and the results are below.

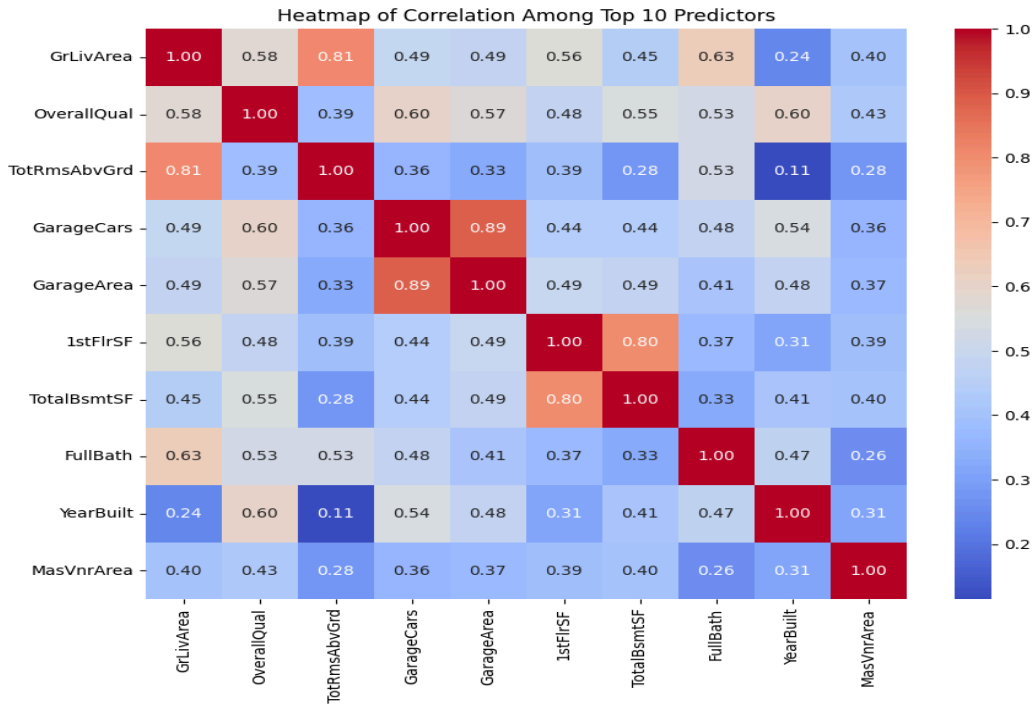


Figure 6: Heatmap illustrating the 10 most correlated variables

After narrowing down the final predictors, a regression was run and the outputs are in the figure below. The following categories were used: GrLivArea, OverallQual, TotRmsAbvGrd, TotalBsmtSF, YearBuilt, and MasVnrArea.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              SalePrice   R-squared:                       0.437
Model:                            OLS   Adj. R-squared:                  0.436
Method:                 Least Squares   F-statistic:                     376.6
Date:                Wed, 05 Mar 2025   Prob (F-statistic):               0.00
Time:                        04:52:36   Log-Likelihood:                -35288.
No. Observations:                2919   AIC:                         7.059e+04
Df Residuals:                    2912   BIC:                         7.063e+04
Df Model:                           6
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
intercept     -3.177e+05   6.45e+04     -4.923      0.000   -4.44e+05   -1.91e+05
GrLivArea        35.4450      3.152     11.246      0.000      29.265      41.625
OverallQual    7841.9836    885.163      8.859      0.000    6106.374    9577.593
TotRmsAbvGrd   3041.5347    879.837      3.457      0.001    1316.370    4766.700
TotalBsmtSF      16.8508      2.271      7.421      0.000      12.398      21.303
YearBuilt       181.6502     33.818      5.371      0.000     115.340     247.960
MasVnrArea       13.1640      5.166      2.548      0.011       3.034      23.294
==================================================================================
```

Figure 7: Model 1 regression results

After finishing the regression for the six categories, the second model was performed. The second model consisted of all categories. To narrow these variables down, our model followed the same rules as model 1. If the P Value was >= 0.05 it was labeled insignificant. After this was done there were 38 total variables. The results for the regression of model 2 is in the figure below.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              SalePrice   R-squared:                       0.514
Model:                            OLS   Adj. R-squared:                  0.507
Method:                 Least Squares   F-statistic:                     80.04
Date:                Wed, 05 Mar 2025   Prob (F-statistic):               0.00
Time:                        04:54:29   Log-Likelihood:                -35074.
No. Observations:                2919   AIC:                         7.023e+04
Df Residuals:                    2880   BIC:                         7.046e+04
Df Model:                          38
Covariance Type:            nonrobust
==========================================================================================
```

Figure 8: Model 2 regression

Furthermore, to evaluate both models the actual and predicted values were reported for both model 1 and 2. With these numbers, the figures below are scatter plots which demonstrate the differences between model 1 and 2. These scatter plots below are comparing the actual sale price vs the predicted sale price. The true value is listed with a red line while a trendline is labeled with a blue line.

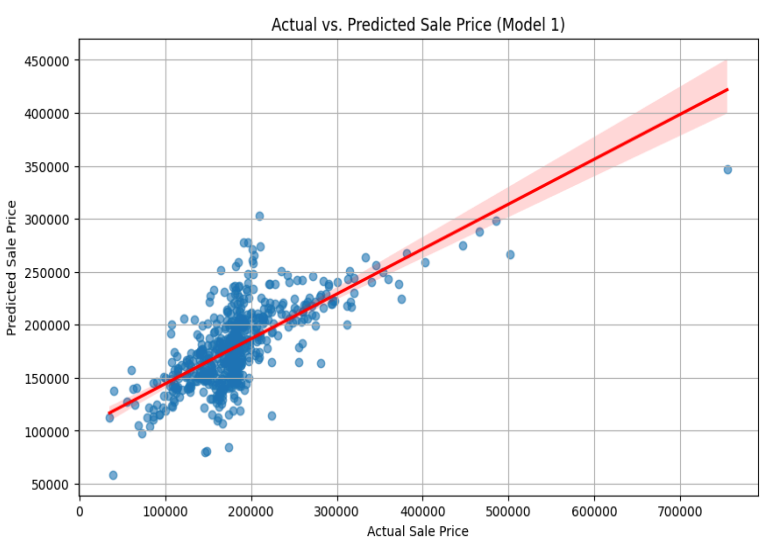


Figure 9: Model 1

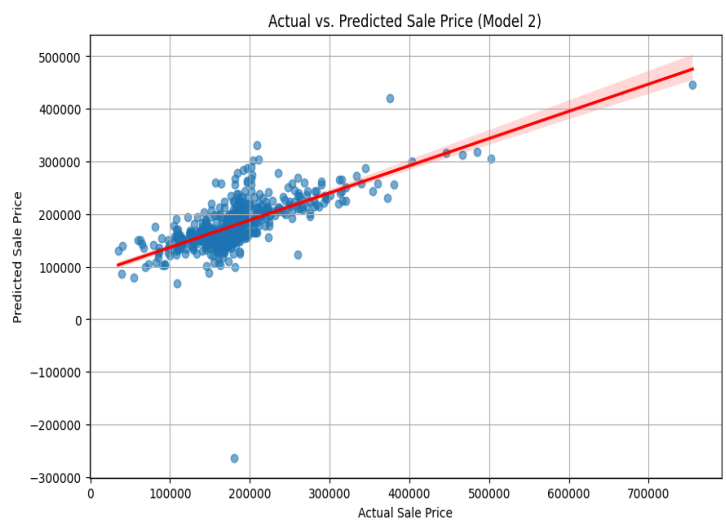


Figure 10: Model 2

IV. RESULTS

In this section, we will take a look at the results of our code and compare our regression model 1 to regression model 2 and interpret the results of the preferred model.

Model Comparison

Key metrics that help determine which model performed better:

- R-squared: Model 2 has a higher r-squared value of 0.514 in comparison to model 1's (figures 5 & 6) r-squared value of 0.437, which suggests that model 2 experiences a larger variance and is a stronger fit. A higher r-squared value and stronger fit are favorable because it shows that the model has a better relationship with the predictors and data.
- Mean squared error (MSE): During our analysis we found that model 1 experiences a lower MSE with a 1,979,519,936.03 in comparison to model 2 which has a 2,029,237,420.21 MSE. A lower MSE is better because it suggests that a model has less errors during its prediction. This suggests that model 1 had less errors and a better predictive accuracy between the predicted and actual sale price.

Model 1 vs. Model 2:

In this analysis, we found that model 1 has a lower r-squared value but a lower MSE, meaning that its predictions are more stable even though it experiences less variability. Model 2 has a higher r-squared value but a higher MSE, meaning that it experiences higher variability and a higher average prediction error. In these 2 key metrics:

- R-squared (variance): Model 1 = 0.437 (43.7%) and Model 2 = 0.514 (51.4%)
- Mean squared error (MSE): Model 1 = 1,979,519,936.03 and Model 2 = 2,029,237,420.21

The choice of the preferred model depends on the objective, as model 1 is best at minimizing prediction error and model 2 has better explanatory power. Model 2 experiences a 51.4% relationship with the saleprice in contrast to model 1's 43.7%. However, model 1's MSE is slightly lower in comparison to model 2, indicating that its predictions have fewer errors.

Interpretation of results:

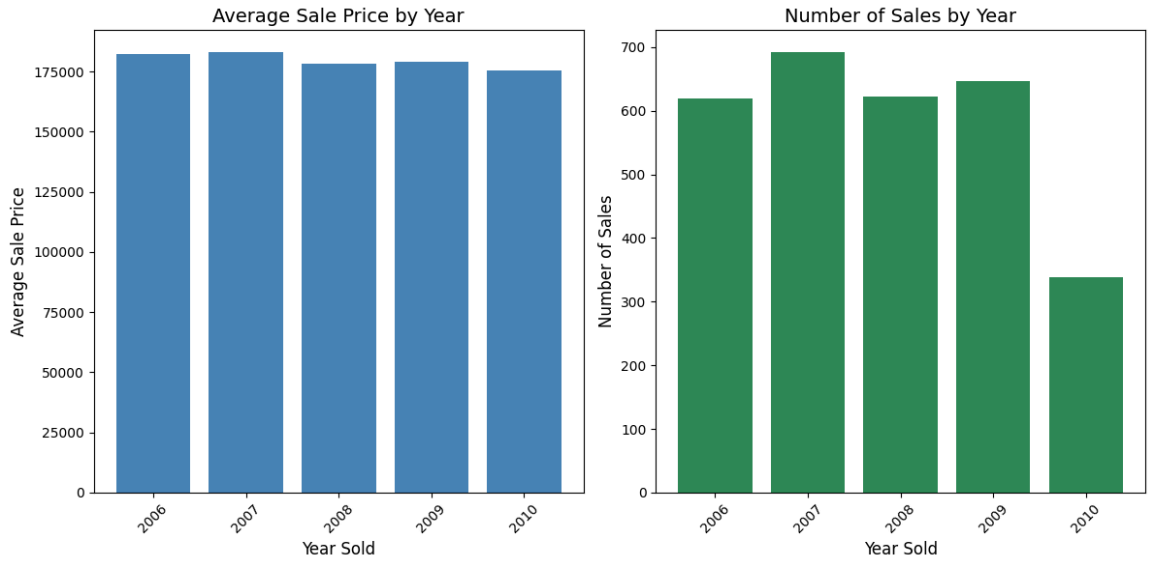


Figure 11 & 12: Bar charts for average sale price and number of sales in a year

These charts help us find and understand the price trends throughout the year. In this we can visualize the average sale price and number of sales during the year.

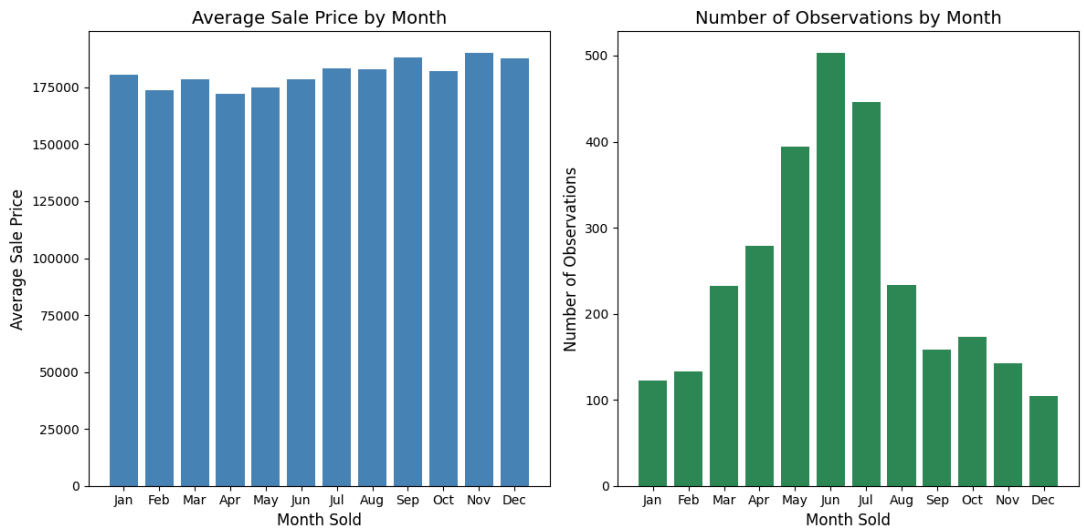
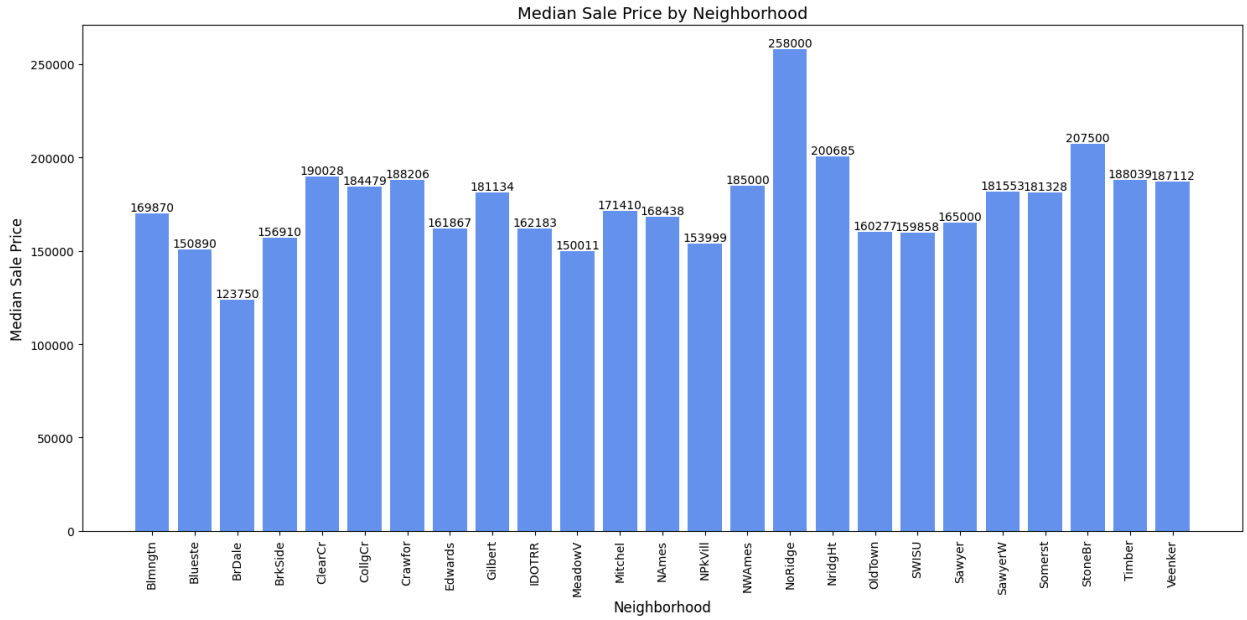


Figure 13 & 14: Bar charts for average sale price and number of observations by month

These charts help us find and calculate the seasonal price trends during the month. Here we can find the average sale price and number of observations throughout the month.



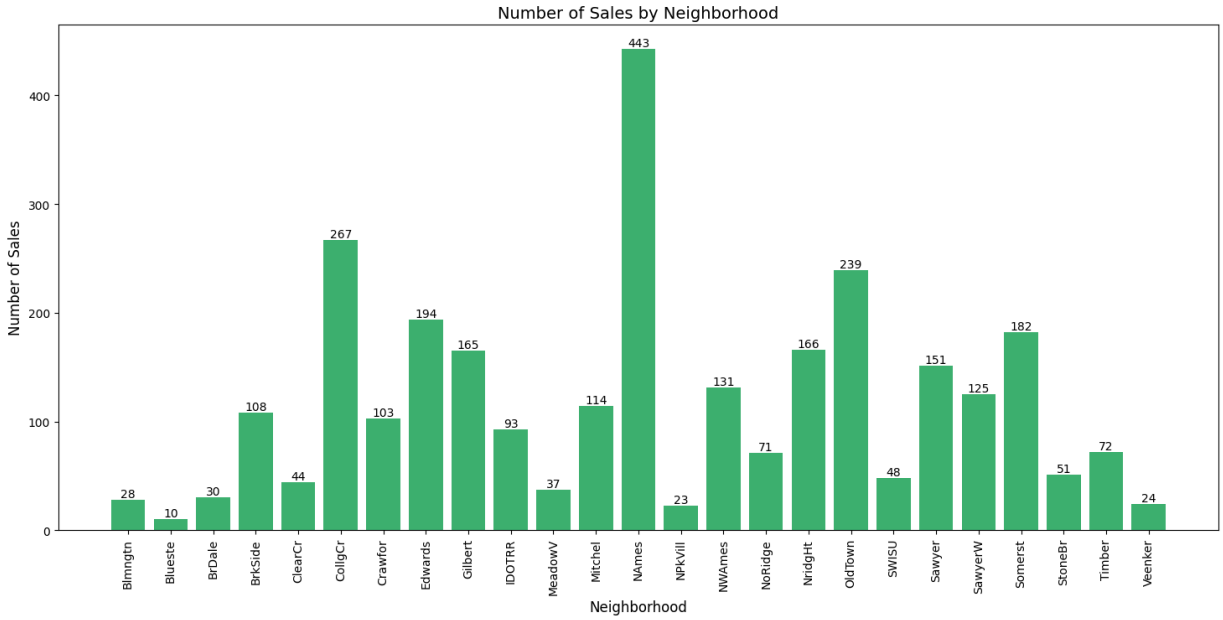
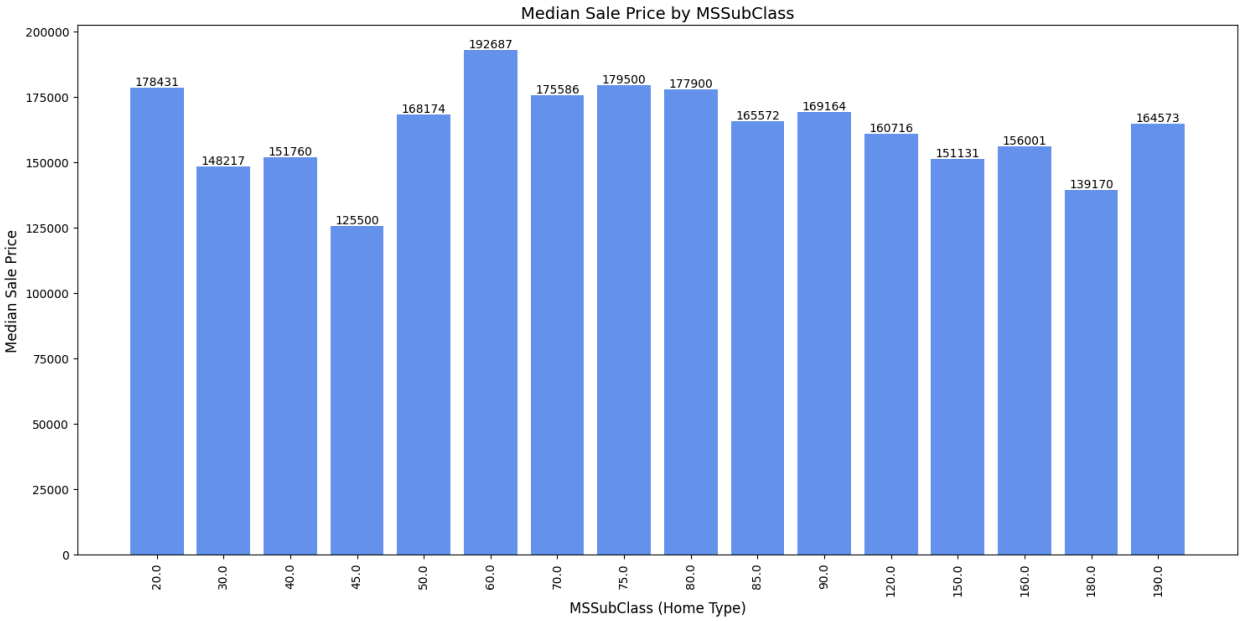


Figure 15 & 16: Box chart of median sale prices and number of sales by neighborhood

This chart helps us find and show the number of sales and median sale price by neighborhood. This helps us understand which neighborhoods tend to have a higher property value.



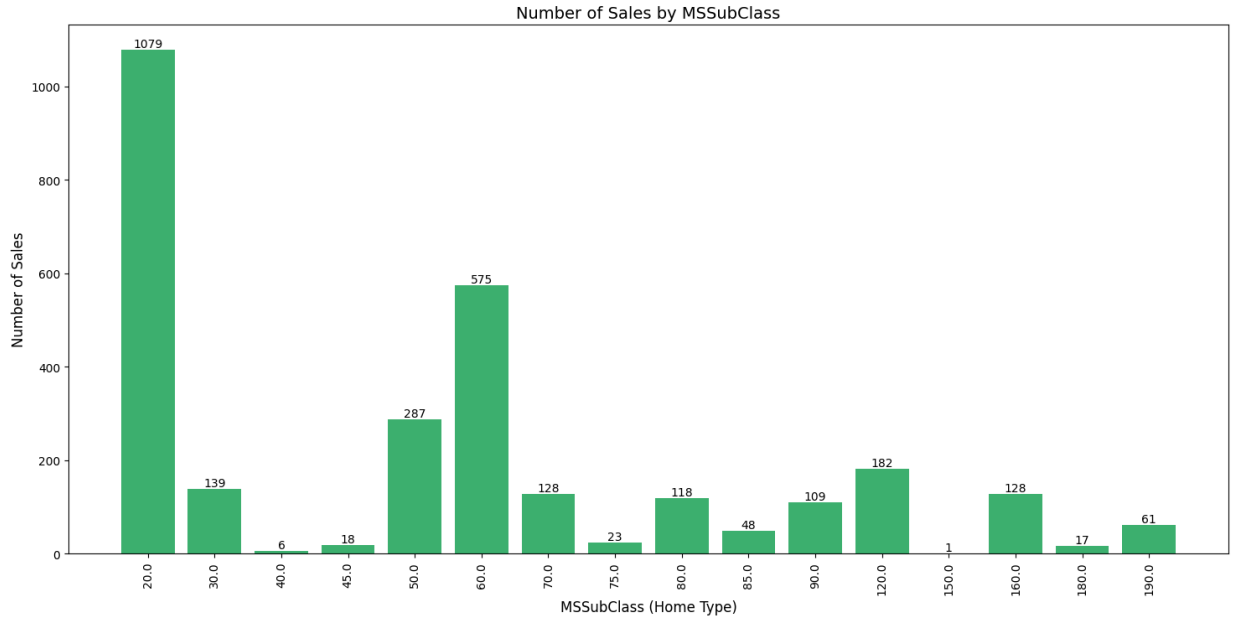


Figure 17 & 18: Box chart of median sale price and number of sales by MSSubClasses

This chart helps us find and visualize the number of sales and median sale price per MSSubClass. MSSubClass represents different types of dwelling styles, which helps us compre pricing across home types.

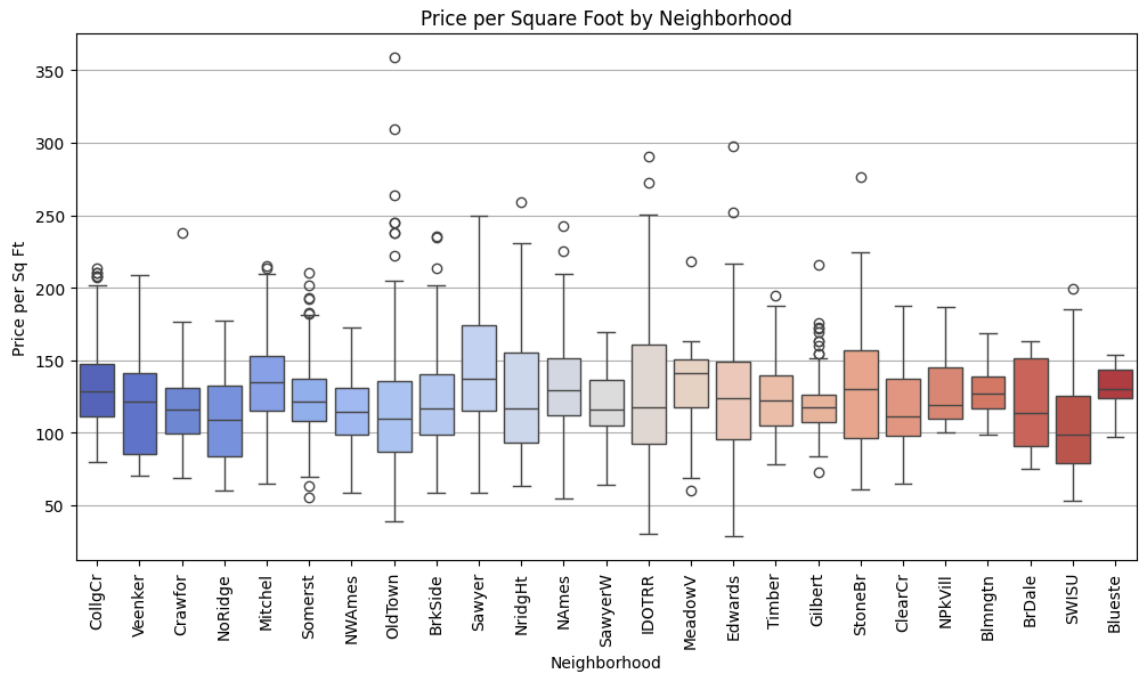


Figure 19: Box plot of price per square foot by neighborhood

This graph helps us visualize the price per square feet in relation to the neighborhood. This metric ultimately helps us compare the home values across neighborhoods more effectively.

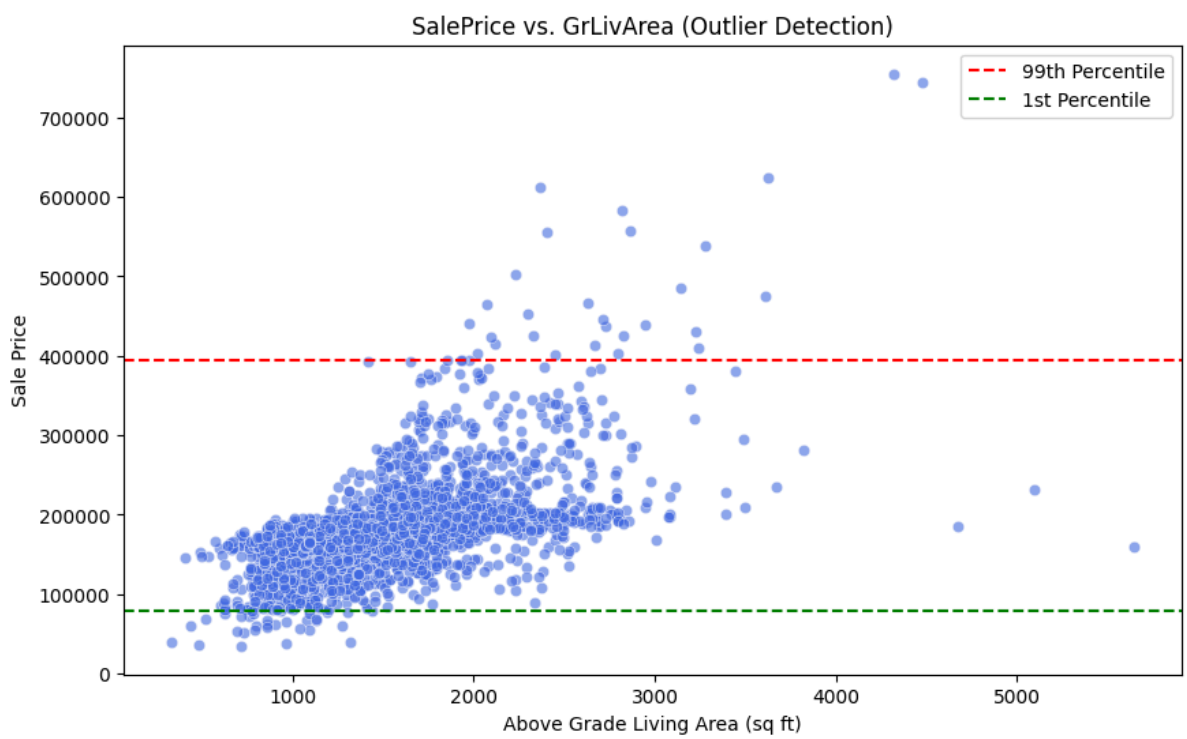


Figure 20: Scatter plot of Saleprice vs. GrLivArea

This scatter plot assists us in analyzing the relationship between SalePrice and GrLivArea. This plot also helps detect outliers and unusual trends in house pricing, by displaying to us the 99th percentile.

Figure 21: Scatter plot of House age vs Saleprice

This scatter plot helps us analyze the relationship between house age and sale price. It also helps us determine whether older homes tend to have higher or lower values.

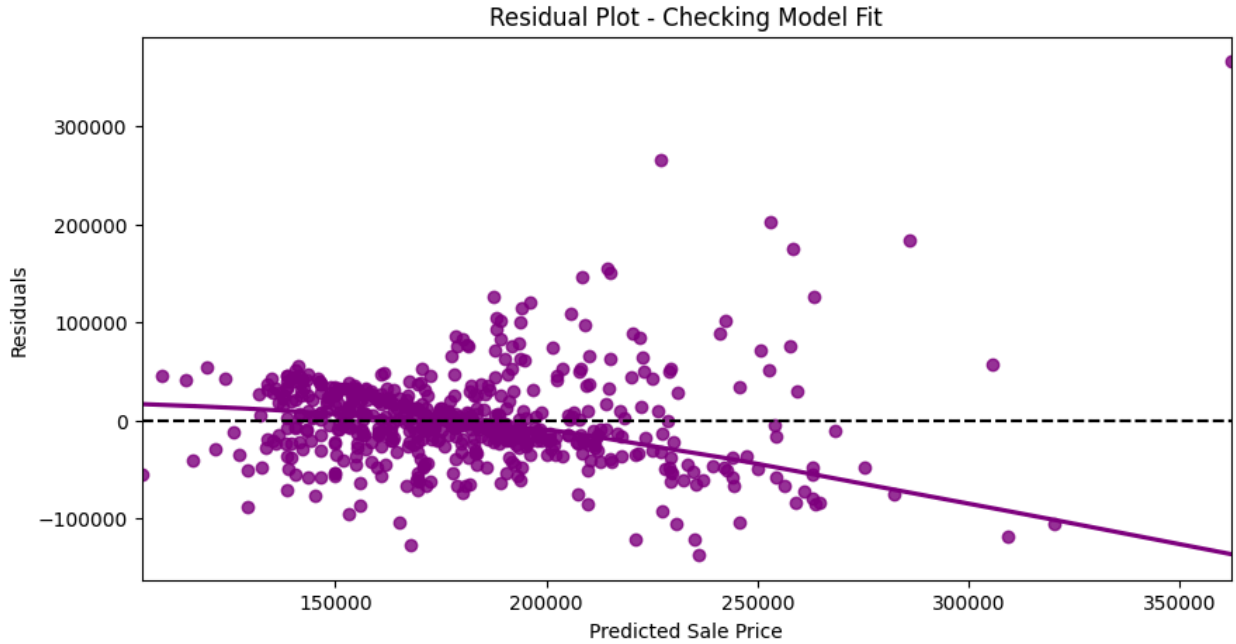


Figure 22: Residual plot for checking model fits

In this graph we are evaluating the accuracy of the predicted sale price. This graph helps us understand how accurately the predicted model fits.

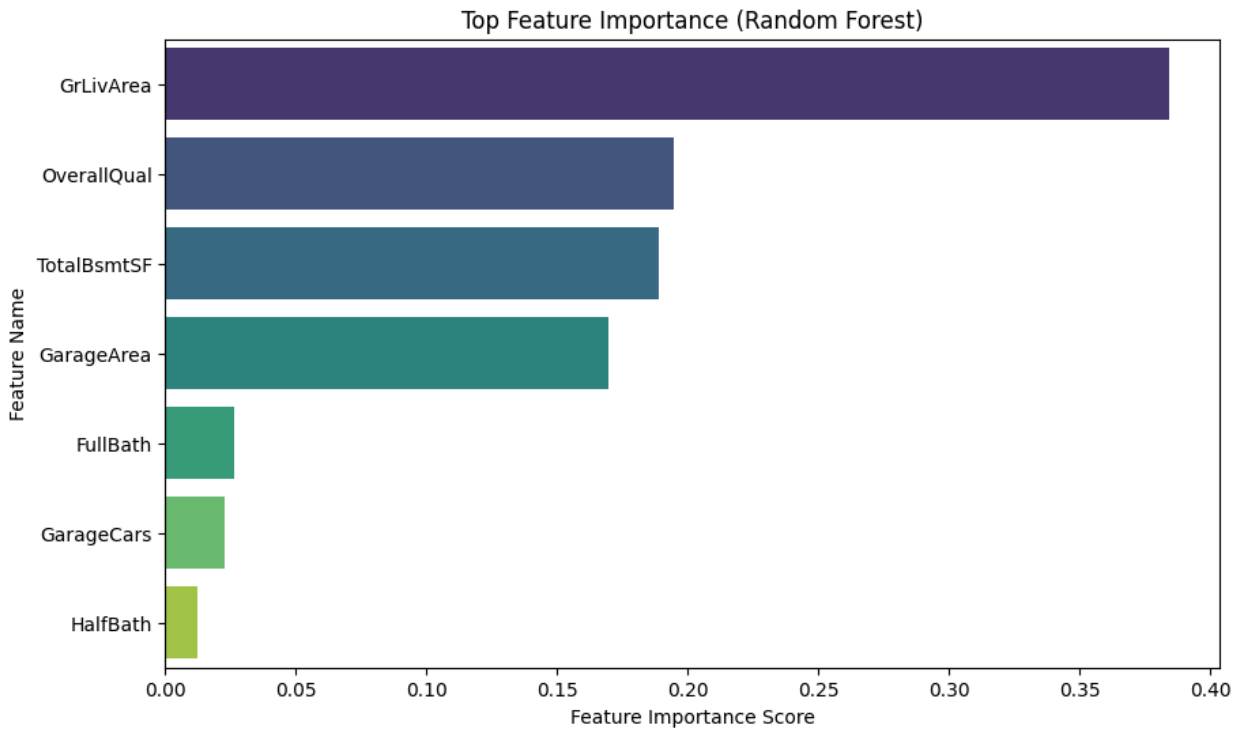


Figure 23: Bar graph for top feature importance

This graph shows us the relationship between the top predictors of SalePrice and their importance scores. This determines the importance of a predictor when figuring out the SalePrice of a house.

V. CONCLUSION

In conclusion, this study applied linear regression techniques to predict housing prices for residential properties sold in Ames, Iowa, between 2006 and 2010. Through analyzing the given data, predictors were identified that specifically influenced the housing market. The study consisted of 2 models, Model 1, which included a smaller set of highly correlated variables, demonstrated a lower mean squared error or MSE, indicating better predictive accuracy. On the other hand, Model 2, which incorporated a larger number of variables, had a higher R-squared value, suggesting a stronger overall fit to the data but at the cost of increased prediction error.

These findings illustrate a powerful data driven strategy that shows performance in the housing market. With this information, buyers, sellers, and agents can use these data driven models to contribute to their decision making. As a home buyer, it can be interpreted that categories of the house can fluctuate the price of the home, this helps the buyer if they were to have a financial budget when buying a home. As a seller, the study illustrates how home improvements such as rooms above ground or home quality drive the price of the property. Finally, as an agent, using these results can help discussion with the customers, both sellers and buyers, on what specifically is driving the price of homes and trends in a certain market.

As a final point, it is important to acknowledge that this study is limited in some aspects. One of which is the data is accurate but it is a small sample size that consists of one town in Iowa from 2006 and 2010. With this, it is important to understand that this might not fully reflect the current housing market.

VI. REFERENCES

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